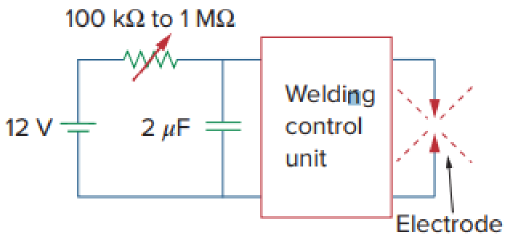
Chapter 7, Problem 86P

Problem

Figure 7.146 shows a circuit for setting the length of time voltage is applied to the electrodes of a welding machine. The time is taken as how long it takes the capacitor to charge from 0 to 8 V. What is the time range covered by the variable resistor?

Figure 7.146:



Step-by-step solution

Step 1 of 2

$$V(t) = V(\infty) + [V(0) - V(\infty)]\, e^{\frac{-t}{\tau}}$$

V(∞) = 12, V(0) = 0

$$V(t) = 12\left[1 - e^{\frac{-t}{\tau}}\right]$$

$$V(t_0) = 8 = 12\left[1 - e^{\frac{-t_0}{\tau}}\right]$$

$$\frac{8}{12} = 1 - e^{\frac{-t_0}{\tau}} \Rightarrow e^{\frac{-t_0}{\tau}} = \frac{1}{3}$$

$$t_0 = \tau \ln(3)$$

Step 2 of 2

For R = 100 K Ω

$$\tau = RC = \left(100\times10^{3}\right)\left(2\times10^{-6}\right) = 0.2\,S$$

$$t_0 = 0.2\,l\,n(3) = 2.197\,S$$

Thus, Similarly where R = 1 M Ω, t0 = 2.197 S.

0.2197 S < t0 < 2.197 sec

Thus,

0.2197 S < t0 < 2.197 sec.